SAMA: Factorized Semantic Anchoring and Motion Alignment for Instruction-Guided Video Editing

Xinyao Zhang, Wenkai Dong, Yuxin Song, Bo Fang, Qi Zhang, Jing Wang et al. | BAIDU

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Research Questions

- How can instruction-guided video editing be factorized to separately handle semantic modifications and motion preservation without relying on external priors?
- Can motion-centric pretext tasks enable a diffusion model to internalize temporal dynamics for better video editing consistency?
- What training strategy allows video editing models to achieve strong zero-shot performance while maintaining competitive supervised results?

Core Idea

Current video editing models struggle with the semantic-motion trade-off: aggressive edits break temporal consistency while motion preservation dilutes semantic changes. SAMA's key insight is to factorize this problem into two separate but complementary capabilities. Semantic Anchoring jointly predicts semantic tokens and video latents at sparse anchor frames to handle instruction-aware structural planning, while Motion Alignment pre-trains on motion-centric restoration tasks (cube inpainting, speed perturbation, tube shuffle) to internalize temporal dynamics from raw videos. This factorization allows the model to learn semantic and motion representations independently during pre-training, then combine them effectively during fine-tuning.

Key Formula

Semantic Anchoring Loss

$$L_{sem} = \|\hat{s} - s\|_1$$

L_sem is the semantic prediction loss, \hat{s} are the projected semantic tokens extracted from N anchor frames, s are the predicted semantic tokens from the model. This L1 loss supervises semantic token prediction to establish reliable visual anchors for instruction-consistent structural planning.

Combined Training Objective

$$L = L_{FM} + \lambda \cdot L_{sem}$$

L is the overall training loss, L_FM is the flow matching loss from Equation 1, L_sem is the semantic anchoring loss, and λ is the loss weight (set to 0.1). This combines standard diffusion training with semantic anchoring supervision.

Key Figure

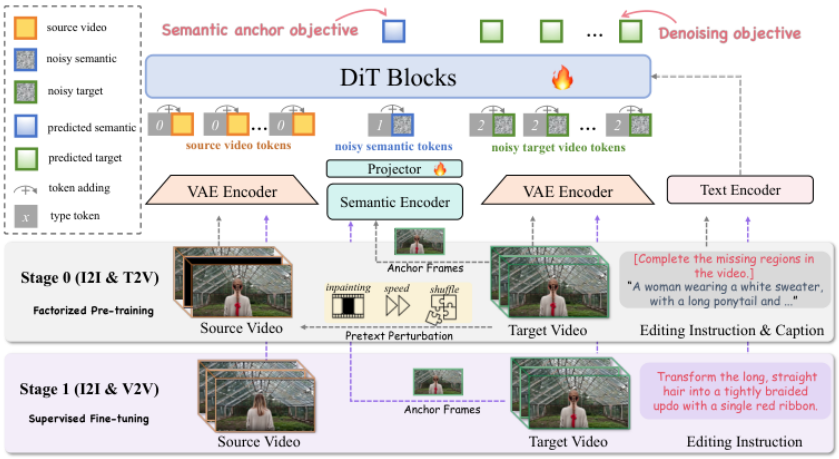

Figure 2: **Overall pipeline.** SAMA first performs factorized pre-training (stage 0) on additional perturbed videos by completing a pretext task conditioned on the given captions. It then performs normal supervised fine-tuning (stage 1) on original source videos. Semantic Anchoring is incorporated in both stages to jointly facilitate semantic representation learning and instruction-guided video editing.

Figure 2: Shows SAMA's complete pipeline including the two-stage training strategy, semantic anchoring mechanism, and motion alignment pretext tasks that are core to the factorized approach.

Limitations

- Zero-shot editing shows temporal inconsistencies in attribute edits (e.g., colors varying across frames)
- Newly added objects may appear blurry in zero-shot setting

- Removal edits can leave residual ghosting artifacts
- Future work needed for long-video editing and fast-motion scenarios with stronger semantic tokenization